

DHANA KARTHIKEYA VENTRAPRAGADA

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EDUCATION

University of California, Berkeley

Dec 2027

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00

Berkeley, CA

- **Relevant Coursework:** Data Structures · Structure & Interpretation of Computer Programs · Machine Learning · Data Engineering · Optimization Models · Principles & Techniques of Data Science · Probability for Data Science

EXPERIENCE

Undergraduate Course Staff (Reader/Tutor)

Aug 2026 – Present

UC Berkeley College of Computing, Data Science, and Society (CDSS)

Berkeley, CA

- Facilitate weekly seminar discussions for **CDSS 198: Careers in Data Science & Computing**, giving feedback on resumes, cover letters, and development plans, and researching/distributing job and internship listings; separately support transfer students in **CDSS 198: Transfer Seminar** by attending both weekly class sessions, holding office hours, and providing on-demand tutoring.

Engineering Intern

May 2026 – Aug 2026

Guang Labs

Remote

- Built an **async video/image domain-classification API** (**FastAPI**, **Auth0**, **Postgres** job store): submit-and-poll **REST** endpoints run classification as **background jobs** with persisted status and granular per-item progress.
- Implemented the Python pipeline end-to-end: resolving media from **4 source types** (local, **S3**, **HTTP**), auto-detecting image vs. short/long video via **ffprobe/ffmpeg**, and classifying via **Gemini** multimodal calls with a trust-gated **resample loop** over fresh windows.
- Split the service into isolated, testable modules behind a central orchestrator, backed by **86 unit, integration, and acceptance tests**; separately built an end-to-end **in-app notification framework** (**Postgres** tables, backend APIs, a **React** dashboard bell) wired into the ingest and export pipelines, with a pull request out for review.

Data Science Intern

Jun 2026 – Aug 2026

IDX Exchange

Remote

- Built a **preprocessing pipeline** for a California home-price model on **70K+ CRMLS single-family records** — data cleaning, statistical outlier caps on lot size and price, and a **chronological train/test split** holding out the latest month.
- Diagnosed and fixed a **train/test data-leakage bug** in the outlier-threshold logic (thresholds were computed over the full dataset), recomputing them from training data only; benchmarked Linear Regression, Decision Tree, Random Forest, and **XGBoost**, with the final **XGBoost** model reaching R^2 **0.88** and **MdAPE 8.64%** on the held-out month.

Undergraduate Course Staff (UCS1)

Jun 2026 – Aug 2026

UC Berkeley Electrical Engineering & Computer Sciences (EECS)

Berkeley, CA

- Course staff for **CS 61A**, an intro CS course taught in **Python, SQL, and Scheme** — ran **4 weekly exam-prep sections** and office hours, built autograders on **Gradescope**, handled regrade/extension requests, wrote and reviewed midterm and final exam questions, and reviewed 2 assignments.

PROJECTS

Movie Genre Classification (Multi-Label NLP) | *Python, PyTorch, SentenceTransformers*

- Built a **multi-label genre prediction system** on **42K+ MovieLens/TMDB films**, parsing nested IMDb metadata with regex, engineering log budget/revenue and temporal features, and embedding title+tagline+overview text via **MiniLM**, **MPNet**, and **e5-large-v2**.
- Benchmarked **Logistic Regression**, **XGBoost**, and **LightGBM** on 1024-dim e5 embeddings; combined a rare-genre cutoff with threshold tuning to address class imbalance, lifting **Macro F1 from 0.48 to 0.66** (Micro F1 0.68) with Logistic Regression beating tree models while training $\sim 12\times$ faster.

Email Spam/Ham Classifier | *Python, scikit-learn, pandas, NumPy*

- Engineered **50+ features** from raw emails — regex counts (links, punctuation, dollar signs), HTML/subject indicators, log-scaled length features, and a curated spam/ham bag-of-words.
- Trained **L1-regularized Logistic Regression** tuned via **GridSearchCV** and validated with 10-fold cross-validation, achieving **92.1% test accuracy** on the held-out leaderboard set.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Scheme

Backend & Infrastructure: FastAPI, Pydantic, PostgreSQL, REST APIs, Auth0, AWS S3, Gemini API, pytest, ffmpeg, React

ML & Data Science: pandas, NumPy, scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, SentenceTransformers

Developer Tools: Git, GitHub, VS Code, IntelliJ, Jupyter Notebook, Google Colab, MATLAB